Proxmox VE 3-Node Enterprise Cluster — Health Audit & Sign-Off

Comprehensive Subsystem Verification, High Availability, Ceph Storage & Network Latency Audit

CLUSTER DOMAIN	NODES AUDITED	AUDIT TIMESTAMP	OVERALL HEALTH
cisai-pve (Version 8.x)	prox022 • prox023 • prox024	October 01, 2026 • 18:20 IST	100% HEALTH_OK (ALL PASS)

Executive Summary: A comprehensive physical and logical cluster health assessment was conducted across all three enterprise hypervisors (`prox022` , `prox023` , `prox024`). The cluster is fully quorate (3/3 votes), Ceph distributed storage is operating at **HEALTH_OK** with all 12 OSDs active and in quorum, ZFS RAIDZ2 pools report zero data corruption, physical 10G interconnect latency is under **0.08ms**, and all 9 production virtual machines are running with zero disruption.

1. Master Subsystem Verification Matrix

Audit Checklist

Subsystem Component	Verification Command	Monitored Target Criteria	Audit Output & Status
Corosync Cluster Quorum	`pvecm status`	Quorate: Yes • Expected: 3 • Total: 3 votes	✔ **PASSED (Ring 1.12a • Config v7)**
Cluster Node Membership	`pvecm nodes`	All 3 nodes online (prox022, prox023, prox024)	✔ **PASSED (3 Nodes Active)**
Ceph Storage Health	`ceph health detail`	HEALTH_OK • Zero slow heartbeats • Zero degraded PGs	✔ **PASSED (HEALTH_OK)**
Ceph Monitor Quorum	`ceph mon stat`	3 MONs active: prox023, prox024, prox022	✔ **PASSED (3/3 MONs In Quorum)**
Ceph OSD Fabric	`ceph osd stat`	12 OSDs up & 12 OSDs in across all 3 nodes	✔ **PASSED (12/12 Up & In • 218 TiB)**
Placement Groups (PGs)	`ceph pg stat`	33 active+clean placement groups	✔ **PASSED (33/33 Active + Clean)**
ZFS Storage Pools	`zpool status zfs-ha`	state: ONLINE • 0 READ / 0 WRITE / 0 CKSUM	✔ **PASSED (127 TiB Clean Across All)**
10G Physical Fabric (nic3)	`ip -br link show nic3`	Carrier: UP • 10Gbps full-duplex • MTU 1500	✔ **PASSED (100% Link Carrier)**
10G Fabric Latency	`ping 192.168.10.x`	Ping response < 0.50 ms with 0% packet loss	✔ **PASSED (0.060 ms – 0.080 ms)**
1G Management (vmbr0)	`ip -br link show vmbr0`	Carrier: UP • Web GUI & SSH active	✔ **PASSED (10.31.6.x Active)**
HA Resource Manager	`ha-manager status`	quorum OK • Master active • All LRMs active	✔ **PASSED (Watchdog Armed)**
Guest Virtual Machines	`qm list`	All 9 guest VMs running (Ceph, ZFS, local-lvm)	✔ **PASSED (9/9 Running • 0 Downtime)**

2. Node-by-Node Physical & Resource Telemetry

Hardware Audit

Node 1: prox022 (10.31.6.172 / 192.168.10.22)

✔ HEALTHY • ONLINE

Processor	AMD EPYC 9454 48-Core Processor (192 CPU Threads)	**Load Average**	0.76, 0.64, 0.67 (Optimal)
Memory Capacity	1.0 TiB Total • 73 GiB Used • **934 GiB Available**	**Swap Usage**	0 B used of 8.0 GiB (0%)
Physical Network	`vmbr0` : UP (10.31.6.172) • `nic3` : UP (192.168.10.22, 10G Storage)	**10G Ping Latency**	**0.079 ms** (0% loss)
Local Storage	ZFS pool `zfs-ha` : ONLINE (0 errors) • `local-lvm` : ONLINE	**Ceph Daemons**	mon.prox022, mgr.prox022 (standby), 4 OSDs
Active Workloads	VM 100 (Windows • 40GB RAM • PID 44173) VM 102 (testdontremove • 20GB RAM • PID 900303) VM 106 (2026090578-jishnua-ubuntu24 • 32GB RAM • 250GB Disk • PID 1958557) VM 9001 (ha-test-ceph-ubuntu2204 • 4GB RAM • PID 318346)		

Node 2: prox023 (10.31.6.173 / 192.168.10.23)

✔ HEALTHY • ONLINE • HA MASTER

Processor	AMD EPYC 9454 48-Core Processor (192 CPU Threads)	**Load Average**	0.93, 0.89, 0.91 (Optimal)
Memory Capacity	1.0 TiB Total • 21 GiB Used • **986 GiB Available**	**Swap Usage**	0 B used of 8.0 GiB (0%)
Physical Network	`vmbr0`: UP (10.31.6.173) • `nic3`: UP (192.168.10.23, 10G Storage)	**10G Ping Latency**	**0.060 ms** (0% loss)
Local Storage	ZFS pool `zfs-ha`: ONLINE (0 errors) • `local-lvm`: ONLINE	**Ceph Daemons**	mon.prox023 (leader), mgr.prox023 (standby), 4 OSDs
Active Workloads	VM 9003 (zfs-benchmark-prod-vm • 4GB RAM • PID 1590485) VM 9004 (vm-9004-test • 4GB RAM • PID 1594063)		

Node 3: prox024 (10.31.6.174 / 192.168.10.24) ✔ HEALTHY • ONLINE • CEPH MGR ACTIVE

Processor	AMD EPYC 9454 48-Core Processor (192 CPU Threads)	**Load Average**	0.72, 0.83, 0.87 (Optimal)
Memory Capacity	1.0 TiB Total • 19 GiB Used • **987 GiB Available**	**Swap Usage**	0 B used of 8.0 GiB (0%)
Physical Network	`vmbr0`: UP (10.31.6.174) • `nic3`: UP (192.168.10.24, 10G Storage)	**10G Ping Latency**	**0.080 ms** (0% loss)
Local Storage	ZFS pool `zfs-ha`: ONLINE (0 errors) • `local-lvm`: ONLINE	**Ceph Daemons**	mon.prox024, mgr.prox024 (active), 4 OSDs
Active Workloads	VM 103 (115-gouthamoshibu-ubuntu24 • 10GB RAM • 150GB Disk • PID 3714007) VM 104 (2026090558-gouthamoshibu-ubuntu24 • 10GB RAM • 150GB Disk • PID 3776614) VM 9002 (zfs-test-prox24-ubuntu2204 • 4GB RAM • PID 1316891)		

3. Ceph Distributed Storage Subsystem

Storage Fabric

The 3-node Ceph storage subsystem is fully synchronized under Ceph Reef with 3-way replication and 0 placement group errors.

```
cluster:
    id:     bcddd50d-c91b-4f6c-b255-19c78e15564f
    health: HEALTH_OK

  services:
    mon: 3 daemons, quorum prox023,prox024,prox022 (age 36m) [leader: prox023]
    mgr: prox024(active, since 40m), standbys: prox022, prox023
    osd: 12 osds: 12 up (since 37m), 12 in (since 6d)

  data:
    pools:   2 pools, 33 pgs
    objects: 3.76k objects, 14 GiB
    usage:   51 GiB used, 218 TiB / 218 TiB avail
    pgs:     33 active+clean
```

4. Corosync Quorum & High Availability

Consensus Fabric

```
Cluster information: cisai-pve • Config Version: 7 • Transport: knet • Secure auth: on
Quorum information: corosync_votequorum • Nodes: 3 • Quorate: Yes • Expected: 3 • Total votes: 3 • Quorum: 2

Membership information:
    Nodeid      Votes Name
0x00000001          1 10.31.6.174 (prox024)
0x00000002          1 10.31.6.172 (prox022, local)
0x00000003          1 10.31.6.173 (prox023)

HA Manager Status:
quorum OK • master prox023 (active) • fencing armed (CRM watchdog active)
lrm prox022 (active) • lrm prox023 (active) • lrm prox024 (active)
```

5. Production Readiness & Sign-Off Authorization

Sign-Off

Role	Designation	Status	Date
Lead Infrastructure Engineer	Nishu Saini • Cloud & Datacenter Ops	**VERIFIED & PASSED**	01-10-2026
Storage & Cluster Lead	CISAI Infrastructure Engineering	**PRODUCTION READY**	01-10-2026
Senior Management Review	Enterprise Infrastructure Review Board	**APPROVED**	01-10-2026